RISK REPORT · DRAFT

Property survey (COPE) - Pacific Coast Logistics

Pacific Coast Logistics · 8500 Port Boulevard, Long Beach, CA 90802
Assessment: Q4 Warehouse Risk Review - Pacific Coast · 2024-11-20 · Assessor: Rachel Torres
Template: Property survey (COPE) v1 · Language: en-GB
Draft: generated by AI, not yet reviewed and issued by an engineer

Executive summary

Summarise the site's risk profile in one or two paragraphs.

Pacific Coast Logistics operates a warehousing and distribution site at 8500 Port Boulevard, Long Beach, California, assessed on 20 November 2024. Warehouse A is a 250,000 sq ft high-bay building of predominantly non-combustible construction (steel frame, metal roof deck and cladding, concrete floor). It holds racked storage to 40 ft on open-frame steel racking with wooden pallets, pallet boxes, cardboard and shrink-wrap packaging. A separate secure, access-controlled, CCTV-monitored and climate-controlled area stores high-value electronics. Flammable liquids are held in IBCs within an FM Approved Minimax containment unit that stands in the open warehouse rather than a separate fire-rated room. The occupancy hazard is rated High because of high-piled mixed combustible storage, poor housekeeping and flammable liquids.

Fire protection is rated only partially adequate. Ceiling and in-rack sprinklers are installed. However, corrosion was seen on branch lines in the south section of Warehouse A. No design standard, density or hydraulic data were provided to confirm adequacy for 40 ft storage. Sprinkler coverage, water supplies and detection and alarm systems could not be verified. In the high-value area there is no water leak detection, despite the client's history of water damage claims. The backup generator for that area was last tested 8 months ago, against a recommended quarterly frequency. Natural hazard exposure (notably earthquake at this Long Beach port location) and external exposures were not assessed from the evidence and remain significant information gaps.

Sources: **E1** Warehouse A - General ✓ “250,000 sq ft warehouse, racked storage to 40ft.” · **E1** Warehouse A - General ✓ “Sprinkler system present but showing signs of corrosion on branch lines in the south section.” · **E2** High-value cargo storage ✓ “No water leak detection sensors installed despite previous water damage claims.” · **E2** High-value cargo storage ✓ “the backup generator for this area was last tested 8 months ago (quarterly testing recommended).” · **E3** Warehouse A - open shelf racking · **E4** Warehouse A - in-rack sprinkler · **E5** Flammable liquid IBC containment

List the most significant risk findings, most important first.

1. Corrosion on sprinkler branch lines in the south section of Warehouse A (250,000 sq ft, racked storage to 40 ft). This may indicate internal deterioration, obstruction or leakage, which would impair protection of high-piled storage. Recommendation priority: High.
2. Sprinkler design adequacy is unverified. No design standard (e.g. NFPA 13 or FM Global Data Sheet 8-9), density or hydraulic calculations were provided for 40 ft mixed commodity storage including plastics. Water supply, sprinkler coverage and detection and alarm systems are also unevidenced.
3. Poor housekeeping, with cardboard and shrink-wrap debris accumulated near the loading docks, adding a readily ignitable fuel load. Recommendation priority: High.
4. No water leak detection in the high-value electronics storage area, despite previous water damage claims. Recommendation priority: High.
5. The backup generator serving the climate-controlled high-value area was last tested 8 months ago, against a recommended quarterly frequency. Recommendation priority: Medium.
6. Flammable liquid IBCs are held in an FM Approved Minimax containment unit, which is a positive feature. However, the unit stands in the open warehouse with no evidenced dedicated suppression, and quantities are unknown.
7. Information gaps: natural hazards (including seismic exposure), external exposures, compartmentation and combustible insulation were not evidenced.

Sources: **E1** Warehouse A - General ✓ “Sprinkler system present but showing signs of corrosion on branch lines in the south section.” · **E1** Warehouse A - General ✓ “Housekeeping is poor — cardboard and shrink wrap debris accumulated near loading docks.” · **E2** High-value cargo storage ✓ “No water leak detection sensors installed despite previous water damage claims.” · **E2** High-value cargo storage ✓ “the backup generator for this area was last tested 8 months ago (quarterly testing recommended).” · **E4** Warehouse A - in-rack sprinkler · **E5** Flammable liquid IBC containment

Overall risk rating for the site.

High

Sources: **E1** Warehouse A - General ✓ “Sprinkler system present but showing signs of corrosion on branch lines in the south section.” · **E1** Warehouse A - General ✓ “Housekeeping is poor — cardboard and shrink wrap debris accumulated near loading docks.” · **E2** High-value cargo storage ✓ “No water leak detection sensors installed despite previous water damage claims.”

Location and external exposures

Describe the location and surroundings (neighbouring occupancies, access, distances).

Insufficient evidence. Needs: The assessment records only the site address (8500 Port Boulevard, Long Beach, CA 90802). The collected evidence covers internal conditions of Warehouse A and the high-value cargo area only. To describe the surroundings, we need a site plan or aerial image showing neighbouring occupancies and separation distances to adjacent buildings. We also need details of road access and fire service access routes, and the distance to the nearest public fire station.

Which external hazards affect the site (neighbouring buildings, flood, earthquake, windstorm, arson)?

Insufficient evidence. Needs: No evidence addresses external hazards. The following are needed: a survey of neighbouring buildings and their occupancies, with construction type and separation distances; the FEMA flood zone designation or a flood map for the site; a seismic hazard assessment, such as USGS ground-motion data, together with any building seismic evaluation or rack anchorage details; windstorm exposure data; and an assessment of site security and arson risk, covering perimeter fencing, lighting, out-of-hours security and external combustible storage. Water leak detection is absent in the high-value area (E2), but this is an internal issue rather than an external exposure.

Overall level of external exposure.

Insufficient evidence. Needs: An overall external exposure rating cannot be assigned without evidence on the following: neighbouring occupancies and separation distances, flood zone status, seismic hazard, windstorm exposure and arson or security risk at 8500 Port Boulevard, Long Beach.

Construction

Describe the building construction - frame, walls, roof, floor area and height.

Warehouse A at 8500 Port Boulevard, Long Beach, has a floor area of approximately 250,000 sq ft, with racked storage to 40 ft, implying a high-bay single-storey building. Photographs show a steel portal frame with open steel roof trusses and purlins supporting a profiled metal roof deck with rooflights. The external walls appear to be metal-faced cladding panels on steel or concrete columns, and the floor is a concrete slab. No information was provided on the year of construction, eaves height, wall or roof panel core material, or roof covering specification.

Sources: **E1** Warehouse A - General ✓ “250,000 sq ft warehouse, racked storage to 40ft.” · **E3** Warehouse A - open shelf racking · **E5** Flammable liquid IBC containment

Predominant construction class.

Non-combustible

Sources: **E3** Warehouse A - open shelf racking · **E5** Flammable liquid IBC containment

Are combustible insulated panels or combustible roof insulation present?

Insufficient evidence. Needs: Manufacturer specification or as-built drawings for the wall cladding and roof panels, confirming the insulation core type (e.g. mineral wool, PIR or EPS) and any FM approval or listing, or a photograph of an exposed panel edge or core.

Describe fire compartmentation and separation between areas.

Insufficient evidence. Needs: The evidence describes a separate secure area for high-value electronics but gives no fire rating for its enclosure. Needed: a site plan showing fire walls and compartments, the fire-resistance ratings of walls separating the high-value cargo area and other areas, and details of fire doors and penetration sealing.

Occupancy and storage

Describe the activities carried out at the site.

The site at 8500 Port Boulevard, Long Beach, is used for warehousing and distribution. Warehouse A is a 250,000 sq ft building with high-bay racked storage served by loading docks, where goods are received and despatched. Within the site there is a separate secure area for storing high-value electronics. This area has controlled access, CCTV monitoring and temperature and humidity control, and is backed by a standby generator. Photographs also show stacked unit load devices or cargo handling equipment on the warehouse floor, which is consistent with cargo logistics operations. An FM Approved containment unit, described as being for flammable liquid IBCs, is also present. The evidence describes no manufacturing or processing activities.

Sources: **E1** Warehouse A - General ✓ “250,000 sq ft warehouse, racked storage to 40ft.” · **E1** Warehouse A - General ✓ “cardboard and shrink wrap debris accumulated near loading docks” · **E2** High-value cargo storage ✓ “Separate secure area for high-value electronics storage. Access controlled, CCTV monitored. Temperature and humidity controlled.” · **E5** Flammable liquid IBC containment

Describe storage - method (racks, solid piled, flow-through), heights, commodities and packaging.

Warehouse A uses racked storage up to 40 ft (approximately 12 m) high. The photographs show tall, multi-tier, open-frame steel racking with open mesh or grated shelf decks, laid out in long numbered rows (bays 35 to 43 are visible). In-rack sprinkler branch lines run at several levels, and the rack uprights have yellow base protectors.

Goods are stored on wooden pallets and in painted pallet boxes or crates sitting on wire mesh decking. Packaging includes cardboard and plastic shrink wrap, and debris of both has built up near the loading docks, which shows poor housekeeping.

High-value electronics are held in a separate secure, climate-controlled area. The evidence does not describe:
- the storage method or height in that area;
- the commodity classification, such as the proportion of expanded or unexpanded plastics;
- flue space dimensions;
- any solid-piled or flow-through storage.

Sources: **E1** Warehouse A - General ✓ “racked storage to 40ft” · **E1** Warehouse A - General ✓ “Housekeeping is poor — cardboard and shrink wrap debris accumulated near loading docks.” · **E2** High-value cargo storage ✓ “Separate secure area for high-value electronics storage.” · **E3** Warehouse A - open shelf racking · **E4** Warehouse A - in-rack sprinkler

Which hazardous materials (flammable liquids, aerosols, gases) are present, in what quantities, and how are they stored?

Flammable liquids are present in intermediate bulk containers (IBCs). The evidence shows one dedicated containment unit on the warehouse floor, identified as flammable liquid IBC containment. It is a red steel bunded enclosure branded Minimax Mobile Services, with galvanised steel side panels, and it carries an FM Approved label. It stands in an open area of the warehouse next to stacked cargo equipment and a pipework manifold; it is not in a separate fire-rated room.

The evidence does not record the following:
- the type, flash point or quantity of the flammable liquids;
- the number of IBCs;
- whether the enclosure has its own fire suppression;
- any aerosols or compressed gases.

Combustible packaging (cardboard and shrink wrap) is present throughout and has accumulated near the loading docks. The high-value electronics may contain lithium-ion batteries, but the evidence does not confirm this.

Sources: **E5** Flammable liquid IBC containment · **E1** Warehouse A - General ✓ “cardboard and shrink wrap debris accumulated near loading docks”

Occupancy hazard classification.

High

Sources: **E1** Warehouse A - General ✓ “250,000 sq ft warehouse, racked storage to 40ft.” · **E1** Warehouse A - General ✓ “cardboard and shrink wrap debris accumulated near loading docks” · **E2** High-value cargo storage ✓ “Separate secure area for high-value electronics storage.” · **E5** Flammable liquid IBC containment

Fire protection

How much of the site is protected by automatic sprinklers?

Insufficient evidence. Needs: Sprinkler layout drawings or a site plan confirming which buildings and areas are sprinklered. The evidence shows ceiling and in-rack sprinklers in Warehouse A only. Coverage of the high-value electronics area, loading docks, offices and any other buildings is not documented.

Is the sprinkler design adequate for the occupancy, storage arrangement and commodity?
Adequacy cannot be confirmed and appears doubtful in part.

Warehouse A is a 250,000 sq ft building with racked storage to 40 ft (E1). The photos show ceiling-level sprinkler mains and branch lines, with vertical feeds running up the rack uprights (E3). In-rack sprinklers are fitted beneath the wire-mesh shelf decking at intermediate levels (E4). In-rack protection is appropriate in principle for racking of this height.

The stored commodity includes wooden pallets and painted plastic or metal containers (E4), and cardboard and shrink-wrap debris was found near the loading docks (E1). This points to at least a mixed commodity with exposed plastics. Such a commodity needs a design matched to a recognised storage standard, such as NFPA 13 or FM Global Data Sheet 8-9. No design standard, design density, sprinkler K-factor or hydraulic calculation was provided, so the design cannot be checked against the 40 ft storage height or the commodity class.

Corrosion was observed on branch lines in the south section (E1). This could indicate microbiologically influenced corrosion (MIC) or trapped water, with a risk of pin-hole leaks, blockage and impaired discharge. This is relevant given the client's history of water damage claims.

The flammable liquid IBC is held in an FM Approved Minimax mobile containment unit (E5), which is a positive feature. However, no dedicated sprinkler protection for flammable liquids is evidenced.

Sources: **E1** Warehouse A - General ✓ “250,000 sq ft warehouse, racked storage to 40ft.” · **E1** Warehouse A - General ✓ “showing signs of corrosion on branch lines in the south section” · **E1** Warehouse A - General ✓ “cardboard and shrink wrap debris accumulated near loading docks” · **E3** Warehouse A - open shelf racking · **E4** Warehouse A - in-rack sprinkler · **E5** Flammable liquid IBC containment

Describe the water supply (pumps, tanks, capacity) and whether it meets the demand.
Insufficient evidence. Needs: Details of the sprinkler water supply, including: town main and/or fire pump details (type, rating, driver), storage tank capacity, the latest pump and flow test results, and the hydraulic demand of the ceiling plus in-rack design for comparison. Photos of the pump house and tank would also help.

Describe fire detection and alarm systems and their maintenance and testing.
Insufficient evidence. Needs: Information on the automatic fire detection and alarm system, including: detector type and coverage (especially in the high-value electronics area), sprinkler waterflow and valve supervisory alarms, monitoring by a central station, and the latest inspection and testing records (for example NFPA 72 test reports). The only systems evidenced are CCTV and access control, which are security measures, and leak detection is confirmed absent.

Overall adequacy of fire protection.
Partially adequate

Sources: **E1** Warehouse A - General ✓ “Sprinkler system present but showing signs of corrosion on branch lines in the south section.” · **E3** Warehouse A - open shelf racking · **E4** Warehouse A - in-rack sprinkler · **E5** Flammable liquid IBC containment · **E2** High-value cargo storage ✓ “No water leak detection sensors installed despite previous water damage claims.”

Natural hazards

Describe flood, earthquake, windstorm and other natural hazard exposures, and the mitigation in place.
Insufficient evidence. Needs: The evidence does not address flood, earthquake, windstorm or other natural hazards at 8500 Port Boulevard, Long Beach. Needed: a seismic hazard assessment (e.g. USGS ground-motion data and site soil or liquefaction classification), confirmation of rack anchorage and sprinkler seismic bracing to NFPA 13, FEMA flood zone mapping and site elevation relative to the port and tsunami inundation zones, and the cause of the previous water damage claims noted in E2 (internal leak or external flooding).

Overall natural hazard exposure.

Insufficient evidence. Needs: A rating cannot be assigned without site-specific natural hazard data. Needed: seismic zone and soil or liquefaction data, the FEMA flood zone designation, tsunami inundation mapping and windstorm exposure for the Long Beach site, plus evidence of mitigation such as rack anchorage, sprinkler seismic bracing and flood defences.

Risk improvement recommendations

List the risk improvement recommendations.

High

Investigate and remediate corrosion on sprinkler branch lines, Warehouse A south section

Observation: Warehouse A is a 250,000 sq ft building with racked storage to 40ft. Corrosion was observed on sprinkler branch lines in the south section, which may indicate internal pipe deterioration, obstruction or leakage risk, compromising sprinkler performance in high-piled storage.

Action: Commission a licensed sprinkler contractor to carry out an internal pipe condition assessment (obstruction investigation) of the south section branch lines, replace corroded pipework as required, and confirm the system design remains adequate for 40ft racked storage. Provide the contractor's report and the latest inspection, testing and maintenance records to insurers.

Sources: **E1** Warehouse A - General ✓ “Sprinkler system present but showing signs of corrosion on branch lines in the south section.” · **E1** Warehouse A - General ✓ “250,000 sq ft warehouse, racked storage to 40ft.”

High

Improve housekeeping and remove combustible packaging debris at loading docks

Observation: Housekeeping in Warehouse A is poor, with cardboard and shrink wrap debris accumulated near the loading docks, providing a readily ignitable fuel load in a busy area adjacent to storage.

Action: Implement a documented housekeeping programme with removal of cardboard and shrink wrap at least each shift, provide designated metal waste/recycling containers or balers located away from the building and racking, and include dock areas in routine management inspections.

Sources: **E1** Warehouse A - General ✓ “Housekeeping is poor — cardboard and shrink wrap debris accumulated near loading docks.”

High

Install water leak detection in the high-value electronics storage area

Observation: The secure high-value electronics area has no water leak detection sensors, despite the client's history of previous water damage claims. Electronics are highly susceptible to water damage.

Action: Install a monitored water leak detection system covering the high-value storage area (including near any pipework, roof drains and HVAC units), with alarms transmitted to a constantly attended location, and establish a response procedure. Review the causes of previous water damage claims and address root causes.

Sources: **E2** High-value cargo storage ✓ “No water leak detection sensors installed despite previous water damage claims.”

Medium	**Reinstate quarterly testing of the backup generator serving the high-value area** Observation: The backup generator supporting the temperature- and humidity-controlled high-value electronics area was last tested 8 months ago, against a recommended quarterly testing frequency, so its reliability during a power outage is unverified. Action: Carry out an immediate load test of the generator, then establish a quarterly (minimum) documented testing and maintenance schedule, including fuel checks and automatic transfer switch operation, with records retained for insurer review. Sources: **E2** High-value cargo storage ✓ “the backup generator for this area was last tested 8 months ago (quarterly testing recommended).”

Sources: **E1** Warehouse A - General ✓ “Sprinkler system present but showing signs of corrosion on branch lines in the south section.” · **E2** High-value cargo storage ✓ “No water leak detection sensors installed despite previous water damage claims.”

Evidence gaps (10)

Questions the evidence could not answer, and what would answer them.

Location and external exposures	The assessment records only the site address (8500 Port Boulevard, Long Beach, CA 90802). The collected evidence covers internal conditions of Warehouse A and the high-value cargo area only. To describe the surroundings, we need a site plan or aerial image showing neighbouring occupancies and separation distances to adjacent buildings. We also need details of road access and fire service access routes, and the distance to the nearest public fire station.
Location and external exposures	No evidence addresses external hazards. The following are needed: a survey of neighbouring buildings and their occupancies, with construction type and separation distances; the FEMA flood zone designation or a flood map for the site; a seismic hazard assessment, such as USGS ground-motion data, together with any building seismic evaluation or rack anchorage details; windstorm exposure data; and an assessment of site security and arson risk, covering perimeter fencing, lighting, out-of-hours security and external combustible storage. Water leak detection is absent in the high-value area (E2), but this is an internal issue rather than an external exposure.
Location and external exposures	An overall external exposure rating cannot be assigned without evidence on the following: neighbouring occupancies and separation distances, flood zone status, seismic hazard, windstorm exposure and arson or security risk at 8500 Port Boulevard, Long Beach.
Construction	Manufacturer specification or as-built drawings for the wall cladding and roof panels, confirming the insulation core type (e.g. mineral wool, PIR or EPS) and any FM approval or listing, or a photograph of an exposed panel edge or core.
Construction	The evidence describes a separate secure area for high-value electronics but gives no fire rating for its enclosure. Needed: a site plan showing fire walls and compartments, the fire-resistance ratings of walls separating the high-value cargo area and other areas, and details of fire doors and penetration sealing.

Fire protection	Sprinkler layout drawings or a site plan confirming which buildings and areas are sprinklered. The evidence shows ceiling and in-rack sprinklers in Warehouse A only. Coverage of the high-value electronics area, loading docks, offices and any other buildings is not documented.
Fire protection	Details of the sprinkler water supply, including: town main and/or fire pump details (type, rating, driver), storage tank capacity, the latest pump and flow test results, and the hydraulic demand of the ceiling plus in-rack design for comparison. Photos of the pump house and tank would also help.
Fire protection	Information on the automatic fire detection and alarm system, including: detector type and coverage (especially in the high-value electronics area), sprinkler waterflow and valve supervisory alarms, monitoring by a central station, and the latest inspection and testing records (for example NFPA 72 test reports). The only systems evidenced are CCTV and access control, which are security measures, and leak detection is confirmed absent.
Natural hazards	The evidence does not address flood, earthquake, windstorm or other natural hazards at 8500 Port Boulevard, Long Beach. Needed: a seismic hazard assessment (e.g. USGS ground-motion data and site soil or liquefaction classification), confirmation of rack anchorage and sprinkler seismic bracing to NFPA 13, FEMA flood zone mapping and site elevation relative to the port and tsunami inundation zones, and the cause of the previous water damage claims noted in E2 (internal leak or external flooding).
Natural hazards	A rating cannot be assigned without site-specific natural hazard data. Needed: seismic zone and soil or liquefaction data, the FEMA flood zone designation, tsunami inundation mapping and windstorm exposure for the Long Beach site, plus evidence of mitigation such as rack anchorage, sprinkler seismic bracing and flood defences.

Evidence cited

E1 Warehouse A - General (note)
E2 High-value cargo storage (note)
E3 Warehouse A - open shelf racking (photo)
E4 Warehouse A - in-rack sprinkler (photo)
E5 Flammable liquid IBC containment (photo)